PageRank

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Reputation System

Exists in many social networks from hotel reviews to search.

Example 1: TripAdvisor allows users to rate hotels and restaurants and search for ones in a particular location.

Example 2: Google uses a reputation system to rank web pages.

Naive PageRank

Even the most obsure search can return a lot of results.

Search engine must order the results in a way that is useful to the user.

Example: "Used cars in New York City".

Page Rank is a method for ranking web pages.

Naive PageRank

If A links to B, then A is said to think highly of B.

If many pages link to B, then B's reputation is high.

$$\begin{tikzcd}
A \arrow[r] \arrow[d] & B \arrow[d] \\
C \arrow[r] \arrow[ur] & D \arrow[u, bend right]
\end{tikzcd}$$

Naive PageRank

Let d_v be the number of outgoing links from page v.

Let π_v be the pagerank of page v, to be defined.

Example: $d_C = 2$.

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Naive PageRank

Definition

$$\pi_v = \sum_{u \to v \in E} \frac{\pi_u}{d_u}$$

Constraint: $\sum_{v \in V} \pi_v = 1$ and $\pi_v \geq 0$ for all $v \in V$.

Intuition: if u has an outgoing edge then u contributes $1/d_u$ to the pagerank of v.

Random walk interpretation

One can think of this as a monkey surfing the web.

At each step, the monkey follows a link from the current page to the next page.

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ C \arrow[r] \arrow[ur] & D \arrow[u, bend right] \end{tikzcd}$$

π_B can be interpreted as the fraction of time the monkey spends at B.

$$\pi_B = \frac{1}{2}\pi_A + \frac{1}{2}\pi_C + \pi_D.$$

PageRank's naive computation

Solve a system of linear equations.

$$\begin{tikzcd}
A \arrow[r] \arrow[d] & B \arrow[d] \\
C \arrow[r] \arrow[ur] & D \arrow[u, bend right]
\end{tikzcd}$$

$$\begin{aligned}
\pi_A &= 0 \\
\pi_B &= \frac{1}{2}\pi_A + \frac{1}{2}\pi_C + \pi_D \\
\pi_C &= \frac{1}{2}\pi_A \\
\pi_D &= \pi_B + \frac{1}{2}\pi_C \\
\pi_A + \pi_B + \pi_C + \pi_D &= 1
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\end{aligned}$$

This gives $\pi_A = 0, \pi_B = 1/2, \pi_C = 0, \pi_D = 1/2$.

PageRank's naive computation

The number of equations is the number of nodes n.

$O(n^3)$ time using Gaussian elimination or matrix inversion.

Infeasible for billion of web pages.

We will deal with this later.

Flaws with naive PageRank (PageRank steal)

A webpage cannot force other webpages to link to it.

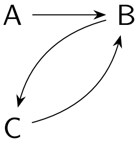
However, it can change its pagerank by changing its outgoing links.

$$\begin{tikzcd}
A \arrow[r] & B \arrow[dl] \\
C \arrow[u] &
\end{tikzcd}$$

In the above, the pagerank of each page is $1/3$.

Let see what happens if C changes its link to A to B.

Flaws with naive PageRank (PageRank steal)



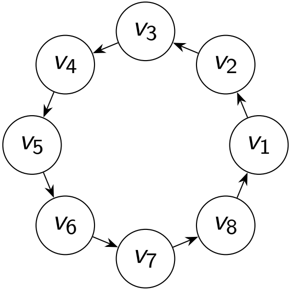
Since A has no link to it, its pagerank is now 0.

The pagerank of B and C is now $1/2$.

Thus, C can "steal" pagerank from A.

Flaws with naive PageRank (PageRank steal)

We can magnify the effect of this by creating a longer cycle.



All pageranks are equal which is $1/8$.

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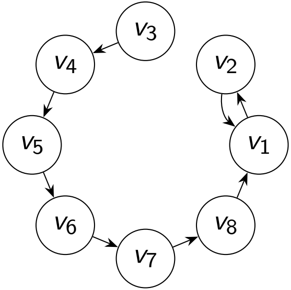

Suppose v_2 redirects its link to v_3 toward v_1. What happens?

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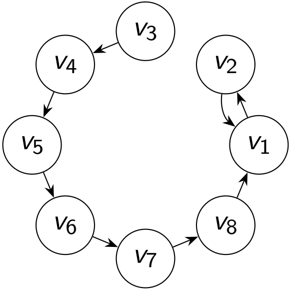

Suppose v_2 redirects its link to v_3 toward v_1. What happens?

The new pageranks of $v_3, v_4, \ldots, v_8$ are now 0. The new pagerank of v_1 and v_2 are now $1/2$ is $1/2$.

Flaws with naive PageRank (Disconnected Components)

An even more serious flaw is that pagerank may not be well-defined.

$$\begin{tikzcd}
A \arrow[r] & B \arrow[dl] & P \arrow[r] & Q \arrow[dl] \\
C \arrow[u] & & R \arrow[u] &
\end{tikzcd}$$

In the above, $\pi_A = \pi_B = \pi_C$ and $\pi_P = \pi_Q = \pi_R$.

Recall: $\pi_A + \pi_B + \pi_C = \pi_P + \pi_Q + \pi_R = 1$.

$\pi_A = \pi_B = \pi_C = \pi_P = \pi_Q = \pi_R = 1/6$ is a solution.

$\pi_A = \pi_B = \pi_C = 0$ and $\pi_P = \pi_Q = \pi_R = 1/3$ is another solution.

Flaws with naive PageRank (Dangling nodes)

An even more serious flaw is that pagerank may not be well-defined.

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In the above, $\pi_A = 0$; $\pi_B = \pi_C/2 + \pi_D/2$; $\pi_C = \pi_D/2$; $\pi_D = \pi_C/2$.

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This implies that $\pi_C = \pi_D = 0$ and hence $\pi_B = 0$. But this violates the constraint that $\pi_A + \pi_B + \pi_C + \pi_D = 1$.

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This means the solution might not even exists.